

Lecture 4 (9/21)

Continuous Distributions

The uniform distribution

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y = np.random.uniform(a, b, size=1000)
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for $y_1 < y_2$ in $[a, b]$.

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- but base on this formula $P(Y = y) = 0$ for all $y \in [a, b]$!
- Therefore, we cannot define a probability distribution by assigning outcomes in the usual way!

Example

The time I wait at the bus stop in minutes is $T \sim \text{Uniform}(0, 5)$.

- What is the probability I wait more than 2 minutes?

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- What is the probability I wait more than 2 minutes?
- What is the probability I wait between 4 and 5 minutes, given that I waited more than 1 minute?

Densities

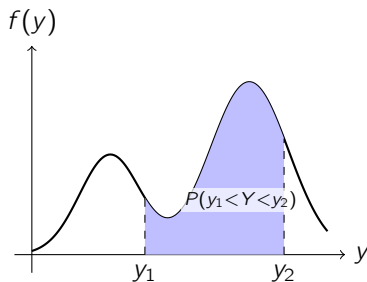
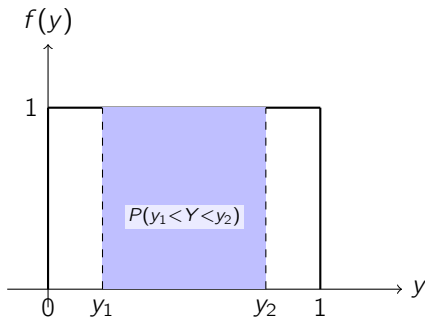
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Densities

A continuous random variable Y has a density $f(y) \geq 0$ with

$$P(a < Y < b) = \int_a^b f(y) dy, \quad \int_{-\infty}^{\infty} f(y) dy = 1.$$

Note $f(y)$ can exceed 1; the only requirement is that the total area under f equals 1.

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- The most important thing for this class is to understand conceptually how probability calculations work, rarely do we evaluate anything but the simplest integral.

Density vs. CDF

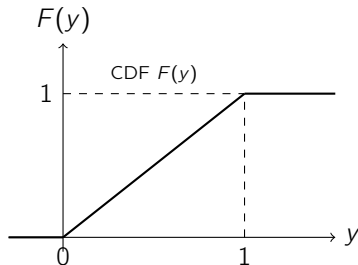
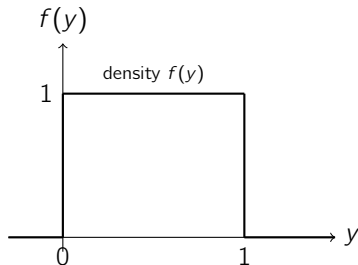
- The density $f(y)$ tells us how probability is spread out: probabilities come from *areas* under f .
- The CDF (cumulative distribution function) accumulates that area from the left:

$$F(y) = P(Y \leq y) = \int_{-\infty}^y f(t) dt \implies f(y) = F'(y)$$

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Example

We often use $\text{Uniform}(0, 1)$ to model an unknown probability. Suppose a soccer team is about to take a penalty kick, and we want to bet on whether they score. We don't know the team's true scoring percentage q , so we model this as

$$\begin{aligned} q &\sim \text{Uniform}(0, 1) \\ Y|q &\sim \text{Bernoulli}(q) \end{aligned}$$

What is the marginal distribution of Y ? What is the density of $q|Y$?

- To find the marginal and conditional note that

$$P(Y = 1|q) = q$$

That is, if we know what q is, the Y is Bernoulli with probability of success q .¹

- The density of q is $f(q) = 1$ when $q \in [0, 1]$ and marginalization works in the same way but with sums replaced with integrals:

$$P(Y = 1) = \int_0^1 P(Y = 1|q)f(q)dq = \int_0^1 qdq = \frac{1}{2}q^2 \Big|_{q=0}^{q=1} = \frac{1}{2}$$

Therefore $Y \sim \text{Bernoulli}(1/2)$

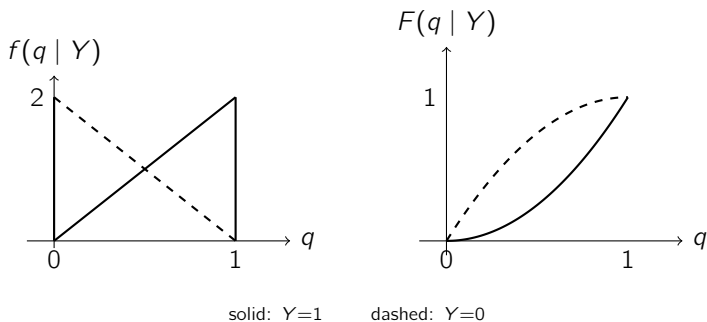
- and using Bayes' (or just the definition of conditional probability)

$$f(q|Y = 1) = \frac{P(Y = 1|q)f(q)}{P(Y = 1)} = 2q$$

¹Technically we are conditioning on a set of measure zero, which has problems, but we will sweep this under the rug

$q|Y$

$$f(q | Y=1) = 2q, \quad f(q | Y=0) = 2(1 - q), \quad q \in (0, 1).$$



Example

Let $U \sim \text{Uniform}(0, 1)$ and find the CDF and densities of U^2 and $\sqrt{U}$.

Example

Let $U \sim \text{Uniform}(0, 1)$ and define $T = -\frac{1}{\lambda} \log(U)$. Find the CDF and density of T .

Solutions

Bus stop, $T \sim \text{Uniform}(0, 5)$:

- $P(T > 2) = 3/5 = 0.6$
- $E[T \mid T > 3] = 4$ (since $T \mid T > 3 \sim \text{Uniform}(3, 5)$)

$X \sim \text{Bernoulli}(1/4)$, $Y \mid X \sim \text{Uniform}(0, 1 + X)$:

- $E[Y] = 5/8 = 0.625$
- $P(Y = 1 \mid X > 1/2) = 0$ (continuous, so $P(Y=y) = 0$ for any y)

Soccer penalty kick, $q \sim \text{Uniform}(0, 1)$, $Y \mid q \sim \text{Bernoulli}(q)$:

- Marginal: $Y \sim \text{Bernoulli}(1/2)$
- $f(q \mid Y=1) = 2q$
- $f(q \mid Y=0) = 2(1 - q)$

$U \sim \text{Uniform}(0, 1)$:

- U^2 : CDF = $\sqrt{v}$, density = $\frac{1}{2\sqrt{v}}$ on $(0, 1)$
- $\sqrt{U}$: CDF = w^2 , density = $2w$ on $(0, 1)$
- $T = -\log(U)/\lambda$: CDF = $1 - e^{-\lambda t}$, density = $\lambda e^{-\lambda t}$, i.e. $T \sim \text{Exponential}(\lambda)$